Cart-Pole Stabilisation in NVIDIA Isaac Sim: Heuristic, Optimal, and Learned Control — A Comparative Study

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Work in Progress

***Abstract*—This report presents a systematic study of cart-pole stabilisation implemented within NVIDIA Isaac Sim. Three control schemes are implemented and evaluated: Proportional-Integral-Derivative (PID), Linear Quadratic Regulator (LQR), and Proximal Policy Optimisation (PPO). Controllers are benchmarked under varying initial conditions and evaluated across a standardised set of performance metrics, including transient effort, steady-state cart regulation, and success rate across initialisation tiers.**

A curriculum learning strategy is developed to extend the PPO agent from near-upright stabilisation towards large-angle recovery, progressing through two completed stages covering initialisation ranges up to 30°. The Stage 2 policy achieves reliable balance across the full angular range $[-180^\circ, 180^\circ]$ in post-hoc evaluation, despite never being trained beyond 30°. Two unexpected observations are documented and form the basis of planned ablation studies: a degradation in training performance with increasing episode duration, and spontaneous generalisation beyond the training distribution that may be attributable to the cosine reward formulation.

This document reflects work completed to date; ongoing extensions are outlined in Section VI.

***Index Terms*—cart-pole stabilisation, PID control, linear quadratic regulator, proximal policy optimisation, reinforcement learning, NVIDIA Isaac Sim, Isaac Lab, curriculum learning**

I. Introduction

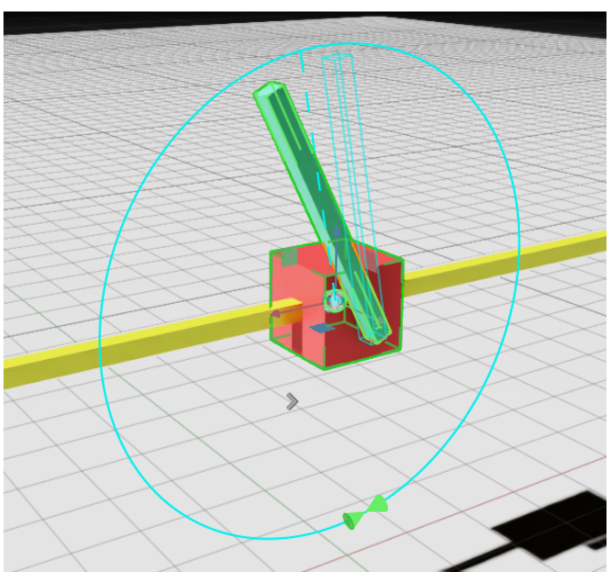

Fig. 1: Cart-pole system as constructed in NVIDIA Isaac Sim.

Simulation environments have become an indispensable substrate for robotics and control research, enabling rapid prototyping of algorithms without the cost and risk of physical hardware. Three platforms are particularly relevant to this work. Gazebo [1] is a widely adopted open-source simulator tightly integrated with ROS/ROS 2, prioritising ecosystem breadth and hardware-in-the-loop compatibility. MuJoCo [2] has become a de-facto benchmark for continuous-control RL research, owing to its accurate convex-optimisation-based contact solver and compact Python interface; GPU-accelerated massively parallel simulation was added in MuJoCo 3 via the MuJoCo XLA (MJX) module [3], making it hardware-agnostic across Nvidia and AMD GPUs, Apple Silicon, and Google TPUs [4]. NVIDIA Isaac Sim [5], used in this work, is built on the Omniverse platform and exposes PhysX 5 for GPU-native rigid-body and articulation dynamics; its RL workflow layer, Isaac Lab [6], enables thousands of parallel environment instances on a single GPU and couples them with RTX path-traced rendering, a full sensor-simulation suite (RGB-D, LiDAR, IMU, contact), and scripted domain randomisation via Omniverse Replicator—features that are particularly valuable for vision-based and sensor-rich policy learning. The principal trade-offs relative to MuJoCo are an exclusive dependency on Nvidia hardware and a substantially larger software stack; a cross-platform comparison using MJX-based environments is identified as a direction for future work (Section VI).

The system studied throughout this work is the cart-pole: a rigid pole pinned at a pivot atop a cart that is free to slide along a horizontal track, driven by a single horizontal force. Despite its mechanical simplicity, the cart-pole distils the essential difficulties of modern control and robot learning into a single, well-defined problem—a quality that has made it the canonical benchmark of the field since Barto, Sutton, and Anderson [7] used it to demonstrate the first neuronlike adaptive controller over four decades ago.

The system's enduring relevance stems from a precise combination of properties that place it at the boundary between tractable and non-trivial. It is open-loop *unstable*: the pole falls without intervention, making active control obligatory rather than merely desirable. Near the upright equilibrium the equations of motion admit a clean linearisation [8], so that classical optimal methods such as LQR yield an exact, analytically grounded solution and a rigorous baseline against which any learning-based approach must be justified. Yet the system is globally *nonlinear*: the small-angle approximation degrades rapidly as the pole departs from vertical, and tasks such as swing-up from the hanging rest position demand a controller that operates coherently across the full state space—a regime where fixed linearisations fail and RL earns its place legitimately. It is also *underactuated*: a single input must simultaneously regulate two mechanically coupled degrees of freedom, making every control decision an inherent trade-off between cart position and pole angle.

The state space is continuous and the dynamics unfold in continuous time, which means the problem belongs firmly to the domain of analogue, time-series control rather than discrete decision-making. The four-dimensional state vector—cart position, cart velocity, pole angle, and pole angular velocity—must be estimated and acted upon at high frequency, placing demands on both the controller architecture and the data pipeline that are representative of real embedded control systems. This makes the cart-pole a natural proving ground for algorithms designed for continuous action spaces, such as LQR, PPO, and SAC, and a stringent test of the sensor and timing fidelity of any simulation environment.

The problem also offers a transparent and principled difficulty ladder. Physical realism can be increased incrementally by introducing dry and viscous friction at the pivot and the track, varying the pole mass and its distribution along the rod, adding external impulsive disturbances, or constraining the cart to a finite track length. Each modification preserves the essential structure of the problem while exposing a new failure mode in controllers designed for the nominal case, allowing a systematic study of robustness that would be opaque in a more complex system. Crucially, the hardware required to validate learned policies on a physical platform is inexpensive and widely available, making sim-to-real transfer—a central challenge in modern robot learning [9]—directly testable without laboratory infrastructure. Extensions to the double and triple pendulum [10] or the rotary (Furuta) pendulum [11] escalate the complexity further, providing a natural research trajectory from the results reported here.

Finally, the cart-pole benefits from four decades of accumulated literature, spanning analytical stability proofs, energy-based swing-up strategies, and a wide range of RL benchmarks, which means that every result obtained here can be situated within a well-understood context and compared against established baselines. For all of these reasons—instability, nonlinearity, underactuation, continuous state and action spaces, controllable difficulty, low-cost physical validation, and rich prior art—the cart-pole is not merely a convenient starting point but a genuinely informative one: a problem small enough to be fully understood and large enough to expose the real differences between control paradigms.

The primary objectives of this work are:

1) Provide a self-contained comparison of PID, LQR, and PPO controllers under a standardised evaluation protocol.
2) Develop a curriculum-based PPO policy capable of progressively larger-angle recovery beyond the linear regime.

Section II describes environment construction. Section III defines state variables, goals, and metrics. Section IV covers PID, LQR, and PPO. Section V reports experimental results, and Section VI outlines ongoing and planned work.

II. Environment Preparation

A. Asset Construction

The cart-pole assembly was constructed directly within the Isaac Sim GUI as a Universal Scene Description (USD) file, the native format of NVIDIA Omniverse, without the use of external modelling tools. The scene comprises three rigid bodies: (i) a rectangular *rail* fixed to the world frame, serving as the sliding track; (ii) a box-shaped *cart* that translates along the rail; and (iii) a rectangular rod *pole* attached to the cart.

B. Physics Configuration

a) Rigid Body and Articulation API.: Physics was applied through Isaac Sim's built-in physics panel. Each body was assigned a `RigidBodyAPI` prim, and the assembly was configured as a single `Articulation`—a robotics primitive that groups a kinematic chain of rigid bodies and joints into a single simulated unit, enabling efficient GPU-parallel state readout and joint-level actuation via the `ArticulationView` API. The rail is attached to the world via a **Fixed Joint**, grounding the entire assembly. The cart is constrained to translate along the rail axis via a **Prismatic Joint**. The pole is connected to the cart via a **Revolute Joint** anchored at the centre of the lateral face of the cart, allowing free rotation in the xz-plane.

b) Joint Parameterisation.: The prismatic (cart) joint is actuated via an `ImplicitActuatorCfg` with effort limit $800\,\mathrm{N}$, velocity limit $100\,\mathrm{m/s}$, stiffness 0, and damping $b_c = 10.0\,\mathrm{N\,s/m}$. The revolute (pole) joint uses a passive actuator with zero effort limit, stiffness 0, and damping b_p. The action space maps a scalar network output to a force command via a scale factor of $100\,\mathrm{N}$ per unit, with forces clipped to $\pm 400\,\mathrm{N}$ at inference. Table I summarises the baseline physical parameters.

TABLE I: Baseline Physical Parameters

Parameter	Symbol	Value
Cart mass	M	$1.0\,\mathrm{kg}$
Pole mass	m	$1.0\,\mathrm{kg}$
Pole length (CoM)	L	$1.23\,\mathrm{m}$
Cart actuator damping	b_c	$10.0\,\mathrm{N\,s/m}$
Pole joint damping	b_p	$0.02\,\mathrm{N\,m\,s}$ (baseline)
Gravity	g	$9.81\,\mathrm{m/s^2}$
Simulation step	Δt	$1/120\,\mathrm{s}$
Control decimation	—	2 (60 Hz effective)
Action scale	—	$100\,\mathrm{N}$
Effort limit	—	$800\,\mathrm{N}$

c) Physics Sanity Check.: The fidelity of the PhysX simulation was assessed qualitatively. An uncontrolled pole released from a small initial angle falls in a manner consistent with the analytical equations of motion, and the LQR controller designed from the linearised model (Appendix A) stabilises the system as predicted by the theoretical gain structure, providing confidence that the simulator reproduces the expected dynamics in the linear regime.

III. State Variables, Goals, and Performance Metrics

A. State Vector

The full state of the system at time t is:

$$\mathbf{s}(t) = \left[x,\ \dot{x},\ \theta,\ \dot{\theta}\right]^\top \tag{1}$$

where:

$$\begin{aligned} x &\triangleq \text{cart position (m)}, \\ \dot{x} &\triangleq \text{cart velocity (m s}^{-1}\text{)}, \\ \theta &\triangleq \text{pole angle from vertical (rad)}, \\ \dot{\theta} &\triangleq \text{pole angular velocity (rad s}^{-1}\text{)}. \end{aligned}$$

B. Task Definition

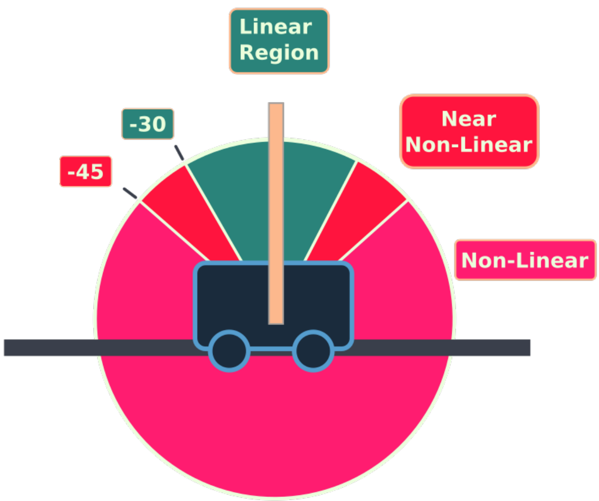

Fig. 2: Cart-pole system and angle regions.

The task is defined as bringing the pole to a stable upright position and maintaining it thereafter, as illustrated in Figure 2. A controller is considered to have achieved balance once the pole enters and remains within:

$$|\theta(t)| \leq \theta_{\text{stable}} \approx 10^\circ \tag{2}$$

for a sustained period following transient motion.

Hard failure is triggered by either:

$$n_{\text{rot}} \geq 3 \quad \text{(full 360° spins accumulated)}, \tag{3}$$

$$|x(t)| \geq 9.0\,\text{m} \quad \text{(cart hits hard rail limit)}. \tag{4}$$

C. Performance Metrics

The following scalar metrics are reported across all evaluation conditions:

1) **Time to first stable state** t_{stable}: elapsed time (s) from episode start until the pole first enters and remains within θ_{stable} for a sustained window. Set to `NaN` on failure.
2) **Survival time** t_{surv}: duration (s) before the first failure condition is triggered, or T if successful.
3) **Transient control effort** $\mathcal{E}_{\text{tr}} = \frac{1}{T_{\text{tr}}}\int_0^{t_{\text{stable}}} u^2(t)\,\mathrm{d}t$: mean squared force during the transient phase (N^2).
4) **Transient smoothness** $\mathcal{S}_{\text{tr}} = \frac{1}{T_{\text{tr}}}\int_0^{t_{\text{stable}}} \dot{u}^2(t)\,\mathrm{d}t$: mean squared control rate during the transient phase (N^2/s^2).
5) **Peak cart displacement** $x_{\text{peak}} = \max_{t \leq t_{\text{stable}}} |x(t)|$: maximum absolute cart excursion during the transient phase (m).
6) **Steady-state RMS cart displacement** $\bar{x}^{\text{ss}}_{\text{rms}} = \sqrt{\frac{1}{T_{\text{ss}}}\int_{t_{\text{stable}}}^{T} x^2(t)\,\mathrm{d}t}$: penalises residual cart drift after stabilisation (m).
7) **Steady-state control effort** $\mathcal{E}_{\text{ss}} = \frac{1}{T_{\text{ss}}}\int_{t_{\text{stable}}}^{T} u^2(t)\,\mathrm{d}t$: mean squared maintenance force after stabilisation (N^2).

D. Evaluation Infrastructure

A dedicated evaluation framework (`EvalCore` / `eval_config.py`) automates the full trial sweep. Each sweep is defined as a `SweepCfg` specifying an angle grid, a controller list, and a maximum episode duration $T_{\max}$. Trials are expanded as the Cartesian product $\{\theta_0\} \times \{\text{controllers}\}$. At the end of a sweep, `EvalCore` writes two CSV artefacts: a per-step *timeseries* file recording $(t,\ x,\ \dot{x},\ \theta,\ \dot{\theta},\ u,\ d_{\text{cart}}, d_{\text{pole}},$ failed), and a per-trial *metadata* file recording outcome, tier, and step count.

a) Initialisation Tier Classification.: Initial angles are partitioned into three tiers:

$$\text{tier} = \begin{cases} \textit{linear} & |\theta_0| \in [0^\circ, 45^\circ) \\ \textit{nonlinear} & |\theta_0| \in [45^\circ, 90^\circ) \\ \textit{full} & |\theta_0| \in [90^\circ, 180^\circ] \end{cases} \tag{5}$$

LQR and PID II are evaluated only in the *linear* tier, consistent with their linearisation assumptions. PPO is evaluated across all three tiers.

IV. Control Strategies

A. PID Controller

a) Definition.: A PID controller computes a control signal $u(t)$ as:

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\,\mathrm{d}\tau + K_d\, \dot{e}(t). \tag{6}$$

Three architectures of increasing sophistication are implemented, all using finite-difference derivatives without filtering.

b) Architecture I: Single-Loop Angle Control.: A single PID loop driven solely by pole-angle error:

$$e(t) = \theta_{\text{ref}} - \theta(t), \quad \theta_{\text{ref}} = 0 \tag{7}$$

with gains $K_p = 300$, $K_d = 70$ (no integral term). Cart position is not regulated and drifts without bound until the track boundary is reached. Architecture I is retained solely as a reference baseline to isolate the contribution of cart-position regulation; *not suitable as a standalone controller.*

c) Architecture II: Dual-Loop Control.: Two independent PID loops run in parallel:

$$u(t) = u_\theta(t) + u_x(t) \tag{8}$$

where the pole loop uses $K_p = 300$, $K_d = 70$, and the cart loop uses $K_p^{\text{cart}} = 30$, $K_d^{\text{cart}} = 15$. The cart correction sign is chosen to reinforce the pole term, simultaneously regulating both degrees of freedom. Architecture II is the primary PID baseline for all multi-controller comparisons.

d) Architecture III: Cascaded PID.: An outer PID on cart position generates a pole-angle setpoint $\theta_{\text{ref}}(t)$ tracked by an inner PD on pole angle:

$$\theta_{\text{ref}}(t) = \text{PID}_x\big(-x(t)\big), \tag{9}$$

$$u(t) = \text{PID}_\theta\big(\theta_{\text{ref}}(t) - \theta(t)\big). \tag{10}$$

Outer loop gains: $K_p = 1.0$, $K_i = 2.5$, $K_d = 3.0$; setpoint clamp $\pm 0.35\,\text{rad}$; integral anti-windup ± 0.25. Inner loop gains: $K_p = 300$, $K_d = 70$. The two-stage latency degrades transient performance, and interaction between the integral term and the plant's underactuation produces residual oscillatory drift. In practice, the cascaded structure exhibited two failure modes not observed in Architecture II: cart displacement frequently exceeded the hard rail limit of $9.0\,\text{m}$ due to unchecked drift driven by the outer integral term, and the loop delay introduced a kickback effect in which the inner pole loop responds to a setpoint already rendered stale by the outer loop, producing abrupt and destabilising force reversals. Architecture III is therefore excluded from the quantitative controller comparison and is retained only as an illustration of the limitations of cascaded integral action in underactuated systems.

e) Sign Conventions (Isaac Sim).: Positive pole angle corresponds to a leftward lean (CCW around y); positive cart position corresponds to a leftward cart displacement; positive force pushes the cart left.

B. Linear Quadratic Regulator (LQR)

a) System Linearisation.: The nonlinear cart-pole equations are linearised about the upright equilibrium ($\mathbf{s}^* = \mathbf{0}$, $u^* = 0$):

$$\dot{\mathbf{s}} = A\mathbf{s} + Bu \tag{11}$$

with A, B derived analytically (Appendix A) using $M = 1.0\,\text{kg}$, $m = 1.0\,\text{kg}$, $L = 1.23\,\text{m}$, $g = 9.81\,\text{m/s}^2$.

b) Optimal Gain Computation.: LQR minimises the infinite-horizon quadratic cost with:

$$Q = \text{diag}(5.0,\ 0.5,\ 100.0,\ 10.0), \quad R = 0.01. \tag{12}$$

The weights were selected manually through iterative tuning. The angle weight ($Q_{33} = 100$) is twenty times the position weight ($Q_{11} = 5$), reflecting the primary objective of pole stabilisation over cart regulation. The velocity weights ($Q_{22} = 0.5$, $Q_{44} = 10.0$) provide damping without overly penalising transient motion. The low effort weight $R = 0.01$ was chosen to permit aggressive actuation, which was found necessary to stabilise the pole within the track limits for larger initialisation angles. The gain vector K_{LQR} is computed once at startup via `scipy.linalg.solve_continuous_are`; forces are clipped to $\pm 500\,\text{N}$.

C. Proximal Policy Optimisation (PPO)

a) Algorithm Overview.: PPO [12] is an on-policy, actor-critic algorithm that maximises a clipped surrogate objective. It is the default RL algorithm in Isaac Lab and provides stable training with relatively few hyperparameter adjustments.

b) Network Architecture.: Both actor and critic use a shared fully-connected MLP trunk with two hidden layers of 64 units and ELU activations, followed by separate policy and value heads. The actor outputs a Gaussian mean via `GaussianMixin`; at inference only the mean is used. Observation dimension: 4; action dimension: 1 (scalar force command). Online observation and value normalisation use a `RunningStandardScaler`.

c) Reward Function.:

$$r_t = w_a \cdot \mathbf{1}_{\text{alive}} - w_t \cdot \mathbf{1}_{\text{term}} + w_p \cdot \cos\theta_t - w_v|\dot{x}_t| - w_w|\dot{\theta}_t| \tag{13}$$

with $w_a = 1.0$, $w_t = 2.0$, $w_p = 2.0$, $w_v = 0.05$, $w_w = 0.02$, and a global reward scale of 0.01. Weights were initialised from the reference Isaac Lab cart-pole implementation and adjusted manually to reflect the task priorities of this work. The termination penalty $w_t = 2.0$ is set to twice the alive bonus to ensure early termination is always penalised relative to survival. The velocity penalties w_v and w_w are kept small to avoid over-constraining the agent during the transient recovery phase. The cosine formulation is preferred over an L_2 angle penalty because it is smooth, bounded, and provides a dense gradient signal across the full angular range, naturally rewarding progress towards upright from any initial angle. A systematic sensitivity analysis of the reward weights is identified as a planned ablation study.

d) Curriculum Learning.: A curriculum progressively expands the initialisation range and tightens termination thresholds. At each episode reset, pole angle is sampled uniformly from $[\theta_{\min}, \theta_{\max}]$ with random sign; cart position from $[-2.0, 2.0]\,\text{m}$; joint velocities from $[-v_{\text{range}}, v_{\text{range}}]$. Two stages are complete; further stages are ongoing (see Section VI).

TABLE II: PPO Curriculum Stages (Completed)

Stage	$\theta_{\min}$ (deg)	$\theta_{\max}$ (deg)	v_r (rad/s)	$\lvert\dot{\theta}\rvert_{\max}$ (rad/s)
1	0	10	0.2	15
2	5	30	0.3	15

e) Training Configuration.: Table III summarises the confirmed training configuration.

V. Results and Analysis

A. Controller Comparison

Tables IV–V summarise controller performance averaged across all trials per controller. PID II and LQR were evaluated over $|\theta_0| \in \{0°, 10°, \ldots, 54°\}$; PPO was evaluated across the

TABLE III: PPO Training Configuration

Parameter	Value
Parallel environments	4,256
Rollouts per update	16
Logged environment steps	160,000
MLP hidden layers	2
MLP hidden units per layer	64
Activation function	ELU
Rollouts per update	16
Learning epochs	8
Mini-batches	8
Discount factor γ	0.99
GAE λ	0.95
Learning rate	3×10^{-4}
LR scheduler	KL Adaptive ($\delta_{\mathrm{KL}} = 0.016$)
Ratio clip ε	0.2
Value clip	0.2
Entropy loss scale	0.02
Value loss scale	2.0
Grad norm clip	1.0
Obs normaliser	RunningStandardScaler
Value normaliser	RunningStandardScaler
Reward scale	0.01
Random seed	42
GPU	NVIDIA RTX 3090

Fig. 3: Representative trajectories: pole angle, cart position, and applied force for each controller under a near-upright initial condition.

Metrics Summary · θ_0 = 28.0° [LINEAR REGIME]

LQR PID_DUAL PPO_INFER

Metric	PID_DUAL	LQR	PPO_INFER
Status	SUCCESS	SUCCESS	SUCCESS
Survival time (s)	20.00	20.00	20.00
Time to stabilise (s)	2.57	2.75	0.82
└ speed ratio [ref: PPO_INFER]	3.14×	3.37×	1.00×
Transient duration (s)	2.55	2.73	0.80
Steady-state duration (s)	17.43	17.25	19.18
— TRANSIENT [t = 0 → t_stable] —			
Peak cart excursion (m)	4.249	3.762	3.453
└ % of track limit (2.5 m)	170.0%	150.5%	138.1%
└ excursion ratio [ref: PPO_INFER]	1.23×	1.09×	1.00×
Ctrl effort ∫u²/T (N²)	502.53	345.50	6168.01
└ effort ratio [ref: LQR]	1.45×	1.00×	17.85×
Ctrl roughness ∫u̇²/T (N²/s²)	598383.32	17160.24	649387.97
└ roughness ratio [ref: LQR]	34.87×	1.00×	37.84×
— STEADY STATE [t_stable → end] —			
RMS cart position (m)	1.7966	3.2976	1.2052
└ % of track limit (2.5 m)	71.9%	131.9%	48.2%
Ctrl effort ∫u²/T (N²)	4.95	0.52	11.81
└ effort ratio [ref: LQR]	9.50×	1.00×	22.66×
Ctrl roughness ∫u̇²/T (N²/s²)	14.53	3.81	1968.22
└ roughness ratio [ref: LQR]	3.81×	1.00×	516.05×

Fig. 4: Performance metrics for all three controllers at $\theta_0 = 28°$ (linear regime). Transient and steady-state metrics are shown for successful trials only.

TABLE IV: Controller Performance (Stabilisation Metrics) — Baseline Physics, N=65 Trials per Controller (LQR, PID II) and N=191 (PPO), $T_{\max}$=20 s

Controller	n_{failed}	$\bar{t}_{\text{stable}}$ (s)	$\theta_{\max}^{\text{stable}}$ (deg)
LQR	8	1.33	44
PID II (Dual-Loop)	3	1.07	49
PPO	0	1.17	180

full angular range $[-180°, 180°]$. Episode duration is fixed at $T_{\max} = 20$ s for all runs. Quality metrics are computed over successful trials only; failed trials contribute solely to n_{failed}.

Key findings from the completed evaluation (Tables IV–V):

- **LQR** demonstrates competitive transient performance within the linear regime, consistent with its optimal gain structure. However, it takes the longest to reach the stable region and produces the highest steady-state cart drift, reflecting the absence of integral action and an inability to correct residual position error once the pole is balanced. Failures increase noticeably as initial angles approach the linearisation boundary.
- **PID II** shows consistent performance across the evaluated range, reaching the stable region faster than LQR with fewer failures. The dual-loop structure provides adequate regulation of both degrees of freedom throughout.
- **PPO** records zero failures across all trials and is the only controller capable of full-range recovery. It achieves the lowest transient effort and tightest steady-state cart regulation. However, steady-state control effort ($\bar{\mathcal{E}}_{\text{ss}} = 43.31\,\text{N}^2$) is approximately 20× higher than LQR ($2.23\,\text{N}^2$). This difference should be interpreted cautiously: PPO is evaluated across the full angular range including large-angle and swing-up initialisations, whereas LQR and PID II are evaluated only in the linear tier ($|\theta_0| \leq 54°$). The elevated effort likely reflects the

TABLE V: Controller Performance (Quality Metrics) — Baseline Physics, successful trials only, $T_{\max}$=20 s. Note that PPO metrics include large-angle and swing-up trials absent from the LQR and PID II evaluation sets; direct comparisons of effort metrics should be treated with caution.

Controller	$\bar{x}_{\text{peak}}$ (m)	$\bar{\mathcal{E}}_{\text{tr}}$ (N^2)	$\bar{x}^{\text{ss}}_{\text{rms}}$ (m)	$\bar{\mathcal{E}}_{\text{ss}}$ (N^2)
LQR	2.69	1581.47	3.18	2.23
PID II (Dual-Loop)	3.43	2374.60	2.00	10.25
PPO	2.86	1245.82	1.26	43.31

energy cost of swing-up recovery from large angles rather than an intrinsic inefficiency of the learned policy. A fair comparison of steady-state effort requires restricting all controllers to the same initialisation range, which is identified as a planned evaluation task.

Taken together, these results suggest a clear performance hierarchy that depends on the operating regime: LQR is the most principled choice within the linear region when analytical guarantees are required; PID II offers comparable or better practical performance with simpler tuning; and PPO is the only viable option beyond the linear regime, at the cost of higher steady-state effort and the absence of formal stability guarantees. Figure 4 illustrates these differences concretely for a representative initialisation of $\theta_0 = 28°$.

B. Curriculum Progression

Functional policies were obtained at Stage 1 ($|\theta_0| \leq 10°$) and Stage 2 ($|\theta_0| \leq 30°$). The Stage 2 policy achieves reliable balance across the full angular range $[-180°, 180°]$ when evaluated post-hoc, despite being trained only on initialisations up to 30°. The primary remaining challenges are reducing steady-state angle error and constraining cart displacement towards the track centre.

Two observations from Stage 2 training and evaluation were not anticipated and are documented here as the basis for planned ablation studies.

a) Episode Duration and Training Performance.: As shown in Figure 5, increasing the maximum episode duration from 480 to 720 steps did not improve and appeared to degrade training performance. The motivation for longer episodes was to provide the agent more time to attempt recovery from difficult initialisations. However, cart-pole stabilisation under Stage 2 conditions is typically achieved within approximately 3 seconds (≈180 steps at 60 Hz). Beyond this point, environments that have already stabilised continue contributing to policy gradient updates without providing a meaningful learning signal, which may smooth the gradient and slow adaptation for environments still experiencing failures. This remains a preliminary observation; the causal mechanism has not been isolated.

Planned ablation: Episode duration will be swept across a range of values under otherwise identical Stage 2 conditions, with mean reward at convergence and convergence speed as primary metrics, to determine whether the observed effect is

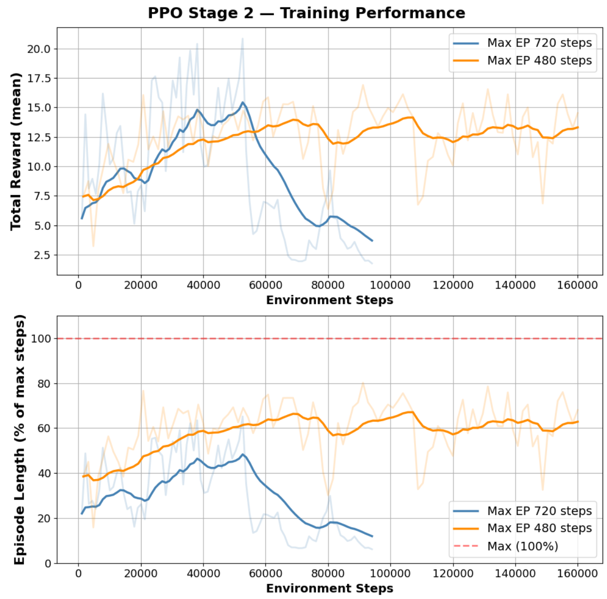

Fig. 5: Effect of maximum episode duration on PPO Stage 2 training performance. The shorter budget (480 steps) produces comparable or higher asymptotic reward than the longer budget (720 steps).

consistent across seeds and how it scales with curriculum stage difficulty.

b) Generalisation Beyond the Training Distribution.: Both Stage 1 and Stage 2 policies were evaluated at initialisation angles beyond their respective training ranges. The Stage 2 policy achieves reliable recovery well beyond 30°, and the Stage 1 policy showed partial recovery up to 160° with reduced consistency. This is possible because the termination conditions were not strictly confined to the linear regime, allowing the agent to encounter and learn from large-angle states incidentally during training. The cosine reward term $r_{\text{pole}} = \cos\theta_t$ likely contributes to this behaviour, as it provides a dense gradient signal across the full angular range rather than only near the upright equilibrium. A related observation is that the policy occasionally applies what appears to be a swing-up strategy at small angles where direct stabilisation would suffice, suggesting the agent may have learned energy-based recovery as a general behaviour.

Planned ablation: Stage 2 training will be repeated replacing the cosine reward term with an L2 angle penalty to assess the contribution of the reward formulation to the observed generalisation. Success rate across a uniform initialisation angle sweep will be the primary comparison metric.

C. Computational Resources

Table VI summarises the average hardware utilisation recorded during Stage 2 training. GPU utilisation remained moderate, suggesting headroom for larger network architectures or increased parallel environment counts in future stages.

TABLE VI: Average Resource Utilization During PPO Stage 2 Training

Metric	Unit	Mean
GPU Utilization	%	65.85
GPU Memory Used	GB	6.99
GPU Temperature	°C	66.92
GPU Power	W	211.61
CPU Utilization	%	13.17
RAM Used	GB	21.01

VI. Ongoing Work and Future Directions

a) PPO Curriculum Completion (Stages 3–5).: Training will be extended to $|\theta_0| \leq 150^\circ$ via three additional stages. An energy-based reward shaping term is implemented and will be gated to activate only when the pole is below the upright threshold, rewarding mechanical energy increase towards the level required for balance. Pole and cart damping will be stage-dependent, gradually reducing towards the underdamped physical regime.

b) SAC Comparison.: PPO will be compared against Soft Actor-Critic (SAC) [13], an off-policy maximum-entropy actor-critic algorithm, under identical training budgets and evaluation conditions.

c) Robustness Evaluation.: Planned disturbance testing includes cart and pole impulses and persistent Gaussian actuation noise. Recovery time t_{rec} will be the primary metric reported across all three controllers.

d) Sensor Noise and Kalman Filtering.: Gaussian observation noise will be injected independently to each state component; a discrete-time Kalman filter [14] will be evaluated as a filtered state estimator for the classical controllers.

e) Physical Parameter Sweeps.: Sensitivity to cart damping, pole damping, and mass ratio will be characterised across all controllers. A domain-randomised PPO policy trained over the swept ranges will be compared to the single-configuration baseline.

f) Double-Pendulum Extension.: The testbed will be extended to a double-pendulum configuration, expanding the state vector to six components $\mathbf{s}(t) = [x,\, \dot{x},\, \theta_1,\, \dot{\theta}_1,\, \theta_2,\, \dot{\theta}_2]^\top$.

g) Cross-Simulator Benchmarking.: The full PID/LQR/PPO pipeline will be reproduced across Gazebo and MuJoCo under a standardised evaluation protocol.

h) Vision-Language Model Integration.: A VLM-based visual stability monitor will be developed as a passive safety layer, trained on rendered Isaac Sim frames labelled as stable, marginal, or failing.

i) Sim-to-Real Transfer.: Deployment onto a physical cart-pole prototype is planned as a later-stage extension, using domain randomisation over mass, cart damping, and pole damping as the primary sim-to-real mitigation strategy.

VII. Conclusion

This work implemented and benchmarked PID, LQR, and PPO controllers for cart-pole balance in NVIDIA Isaac Sim. The dual-loop PID configuration simultaneously regulates pole angle and cart position, serving as the classical baseline against which LQR and PPO are compared. LQR outperformed PID in the linear regime due to its optimal full-state feedback structure, while PPO achieved the highest overall success rate by learning a nonlinear policy from interaction. A two-stage curriculum has been established as the foundation for large-angle recovery, with the Stage 2 policy achieving reliable balance across the full angular range $[-180^\circ, 180^\circ]$, despite being trained only on initialisations up to 30°.

Two unexpected observations emerged during RL-PPO training and evaluation. First, increasing episode duration degraded rather than improved training performance, suggesting that episodes extending well beyond the typical time-to-stabilisation introduce a diluted gradient signal from already-converged environments. Second, the policy generalised substantially beyond its training distribution, achieving reliable recovery across the full angular range without explicit swing-up training. Both observations are documented as preliminary findings and form the basis of planned ablation studies described in Section VI.

The project established a complete development pipeline: asset construction in USD, physics parameterisation via Isaac Lab APIs, three controller implementations, a structured evaluation framework, and curriculum-based PPO training at scale on an RTX 3090. Ongoing work will complete the curriculum through 150°, add robustness and noise evaluation, and extend to a double-pendulum system.

Appendix

The nonlinear equations of motion for the cart-pole are:

$$(M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = u, \tag{14}$$

$$ml^2\ddot{\theta} + mgl\sin\theta + ml\ddot{x}\cos\theta = 0. \tag{15}$$

Linearising about $(\theta, \dot{\theta}, \ddot{\theta}) = (0, 0, 0)$:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{Ml} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{Ml} \end{bmatrix}. \tag{16}$$

With $M = m = 1.0\,\text{kg}$ and $L = 1.23\,\text{m}$, the eigenvalues of A are $\{0,\ 0,\ \pm\sqrt{2g/L}\} \approx \{0,\ 0,\ \pm 3.99\}$, confirming the open-loop instability of the upright equilibrium. The resulting optimal gain vector, computed via `scipy.linalg.solve_continuous_are`, is:

$$\begin{aligned} K_{\text{LQR}} &= \left[k_x,\ k_{\dot{x}},\ k_\theta,\ k_{\dot{\theta}}\right] \\ &= \left[-22.36,\ -32.49,\ -240.26,\ -79.96\right] \end{aligned} \tag{17}$$

where the gain ordering corresponds to the state vector $[x,\, \dot{x},\, \theta,\, \dot{\theta}]^\top$. The large magnitude of $k_\theta = -240.26$ relative to $k_x = -22.36$ reflects the cost weighting $Q_{33} = 100 \gg Q_{11} = 5$, prioritising pole stabilisation over cart position regulation. The negative signs are consistent with the sign conventions adopted in Isaac Sim, where positive force pushes the cart left.

References

[1] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2004, pp. 2149–2154.

[2] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.

[3] Google DeepMind, "MuJoCo 3: A General Purpose Physics Simulator," 2023, version 3.0, released October 2023. MuJoCo XLA (MJX) module introduced GPU/TPU-accelerated parallel simulation via JAX/XLA. [Online]. Available: https://github.com/google-deepmind/mujoco

[4] K. Zakka, B. Tabanpour, Q. Liao, M. Haiderbhai, S. Holt, J. Y. Luo, A. Allshire, E. Frey, K. Sreenath, L. A. Kahrs, C. Sferrazza, Y. Tassa, and P. Abbeel, "MuJoCo Playground," 2025. [Online]. Available: https://arxiv.org/abs/2502.08844

[5] NVIDIA Corporation, "NVIDIA Isaac Sim," https://developer.nvidia.com/isaac-sim, 2023.

[6] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrügg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg *et al.*, "Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning," *arXiv preprint arXiv:2511.04831*, 2025. [Online]. Available: https://arxiv.org/abs/2511.04831

[7] A. G. Barto, R. S. Sutton, and C. W. Anderson, "Neuronlike adaptive elements that can solve difficult learning control problems," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-13, no. 5, pp. 834–846, 1983.

[8] R. V. Florian, "Correct equations for the dynamics of the cart-pole system," Center for Cognitive and Neural Studies, Tech. Rep., 2007.

[9] W. Zhao, J. P. Queralta, and T. Westerlund, "Sim-to-real transfer in deep reinforcement learning for robotics: A survey," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, 2020, pp. 737–744.

[10] K. Furuta, M. Yamakita, and S. Kobayashi, "Swing-up control of inverted pendulum using pseudo-state feedback," *Proc. Inst. Mech. Eng. Part I: J. Syst. Control Eng.*, vol. 206, no. 4, pp. 263–269, 1992.

[11] K. J. Åström and K. Furuta, "Swinging up a pendulum by energy control," *Automatica*, vol. 36, no. 2, pp. 287–295, 2000.

[12] J. Schulman *et al.*, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[13] T. Haarnoja *et al.*, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1861–1870.

[14] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Trans. ASME – J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, 1960.